The documentation and process conversion measures necessary to comply with this amendment shall be completed by 14 July, 2000.

INCH-POUND

MIL-PRF-19500/500C
AMENDMENT 2
14 April, 2000
SUPRESEDING
AMENDMENT 1
6 April 1998

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, TRANSIENT VOLTAGE SUPPRESSOR
UNIPOLAR TYPES 1N5555 THROUGH 1N5558, 1N5907, 1N5629A THROUGH 1N5665A
JAN, JANTX, JANTXV, AND JANS

This amendment forms a part of MIL-PRF-19500/500C, dated 8 July 1997 and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.3. $P_{M(AV)}$; delete "(see 6.3)".

PAGE 2

3.3 Delete and substitute as follows:

"3.3. Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

$I_{(BR)}$ - - - - - - - - - - - -Reverse breakdown current at a specified condition."

PAGE 3

Figure 1, internal construction of drawing, following 6; add "7".

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PAGE 3 Continued

Figure 1, dimension table; delete and substitute as follows:

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BD	.215	.235	5.46	5.97	
BL	.293	.357	7.44	9.07	3
BLT	---	.570	---	14.48	
CD	.045	.100	1.14	2.54	5
LD	.025	.035	0.64	0.89	
LL	1.000	1.625	25.40	41.28	
LU	---	.188	---	4.78	4

“

PAGE 4

Add the following paragraph:

“4.1.1. Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-19500 and as specified herein except, lot accumulation period shall be 6 months in lieu of 6 weeks.”

4.3, screen table, heading; delete “table II” and substitute “table IV”.

4.3, screen 13; delete and substitute as follows:

“

13	Group A, subgroup 2 and 3 electrical parameters not applicable for this screen (performed in screen 12).	Group A, subgroup 2 and 3 electrical parameters not applicable for this screen (performed in screen 12).

“

PAGE 5

4.4.3.1, Group C inspection; C7, conditions column, end of conditions column; add “22 devices, c = 0.”

4.4.3.1, Group C inspection; C8, conditions column, end of conditions column; add “22 devices, c = 0.”

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PAGE 5

* 4.4.3.1 Group C inspection; Add “C5 Not applicable.”

* 4.4.3.1, Group C inspection; C7; Change “C7” to “C8”.

* 4.4.3.1, Group C inspection; C7; Change “C8” to “C9”.

4.5.1, paragraph e; delete and substitute as follows:

“e. Read and record I_D and $V_{(BR)}$ at T_A = +25°C. Devices with ΔI_D > 50 percent (100 percent for steady-state operation life) of the initial reading or 1 μA dc, whichever is greater, or $\Delta V_{(BR)}$ > ±2 percent (±5 percent for steady-state operation life) initial value shall be considered defective. Remove defective devices and record the number of failures. (For steady-state operation life, I_D maximum limit shall be two times the group A limit).”

* 4.5.1, paragraph e; delete the last sentence; “(For steady-state operation life, I_D maximum limit shall be two times the group A limit).”

PAGE 7

*Table I, Subgroup 4, Forward voltage test, Conditions column, delete and substitute “I_{FM} = 100 A (pk), t_p = 8.3 ms (max), duty cycle = 4 pulses per minute (max).

PAGE 8

Table II, step 1, symbol column; delete “I_R” and substitute “I_D”.

PAGES 9 and 10

Table III, characteristics and ratings, column 5; delete “I_R” two places and substitute “I_D” two places.

* Table III, Maximum peak pulse current (I_{PP}), column 7; delete t_p = 1μs and substitute “t_p = 1ms”

PAGE 10

Add table title as follows: ”TABLE III. Characteristics and ratings.”

PAGE 13

* Figure 4, Current impulse waveform, delete the label TIME (t) IN MICROSECONDS and substitute “TIME (t) IN MILLISECONDS.”

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NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5961-2249)

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